Xinkai Shu

Contact Information

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Research Interest

Online Algorithms
Dynamic Graph Algorithms

Education

2019 - 2024 (expected)	**Doctor of Philosophy** Computer Science The University of Hong Kong, Hong Kong, China Advisor: Prof. Zhiyi Huang
2015 - 2019	**Bachelor of Engineering** Computer Science and Technology Institute for Interdisciplinary Information Sciences ("Yao Class") Tsinghua University, Beijing, China Instructor: Prof. Ran Duan

Conference Publications

Online Nash Welfare Maximization Without Predictions
with Zhiyi Huang, Minming Li & Tianze Wei
19th Conference on Web and Internet Economics (**WINE 2023**)

A Randomized Algorithm for Single-Source Shortest Path on Undirected Real-Weighted Graphs
with Ran Duan, Jiayi Mao & Longhui Yin
64th IEEE Annual Symposium on Foundations of Computer Science (**FOCS 2023**)

The Power of Multiple Choices in Online Stochastic Matching
with Zhiyi Huang & Shuyi Yan
54th ACM SIGACT Symposium on Theory of Computing (**STOC 2022**)

Online Stochastic Matching, Poisson Arrivals, and the Natural Linear Program
with Zhiyi Huang
53rd ACM SIGACT Symposium on Theory of Computing (**STOC 2021**)

Manuscripts

Theses

TBD
Doctoral thesis, advised by Prof. Zhiyi Huang, 2024

Quick Algorithms for Dynamic Edge Coloring (in Chinese)
Bachelor thesis, instructed by Prof. Ran Duan, 2019

Honors and Awards

2014 **Gold Medal**
China National Olympiad of Informatics, Shenzhen, Guangdong, China

2013 **Gold Medal**
China National Olympiad of Informatics, Chengdu, Sichuan, China

Invited Talks

A Randomized Algorithm for Single-Source Shortest Path on Undirected Real-Weighted Graphs
CCF Forum for distinguished Ph.D. Candidates in Theoretical Computer Science 2023
The Hong Kong Polytechnic University, Hong Kong, China, July 20-21, 2023

A Randomized Algorithm for Single-Source Shortest Path on Undirected Real-Weighted Graphs
Institute for Theoretical Computer Science (ITCS) Seminar
Shanghai University of Finance and Economics, Shanghai, China, July 11, 2023

Online Nash Welfare Maximization Without Predictions
Complexity & Algorithms Workshop 2023
Shandong University, Jinan, Shandong, China, April 1-2, 2023

The Power of Multiple Choices in Online Stochastic Matching
International Joint Conference on Theoretical Computer Science – Frontier of Algorithmic Wisdom (IJTCS-FAW 2022)
City University of Hong Kong, Hong Kong, China, August 15-19, 2022 (virtually)

Research Visits

Mar. – Aug. 2022	Shanghai University of Finance and Economics Shanghai, China

Teaching Experience

Teaching assistant as PhD student at The University of Hong Kong:

2022 Spring	Design and analysis of algorithms
2021 Fall	Design and analysis of algorithms (Advanced)
2021 Fall	Advanced algorithm analysis
2020 Spring	Design and analysis of algorithms

Additional Information

Language	Native Chinese, Fluent English, Intermediate Japanese
Programming	Proficient in C++, Python, PASCAL
Puzzle	I love solving puzzles, especially mathematical puzzles (e.g. sudoku, kakuro, slitherlink and masyu), as well as chess problems, go problems and tsume-shogi